

Classification of Defects in

Photovoltaic Modules

Computer Vision — Assignment 2

Master's in Electrical and Computer Engineering · FEUP

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Deep learning for thermal IR anomaly detection on solar panels

Motivation

Recent solar-farm infrastructure failures in Portugal and Spain highlight the need for pre-emptive, automated inspection.

Manual expert inspection of large solar farms is slow, costly, and not scalable.

Thermal infrared (IR) imaging from UAVs can reveal hidden defects before they escalate.

Our goal: build AI classifiers that detect and type PV module defects from thermal signatures — without requiring on-site experts.

Project Objectives

Three classification tasks

Binary — anomaly vs. no-anomaly

Multi-11 — 11 defect types (defects only)

Multi-12 — 11 defects + no-anomaly

Evaluation & comparison

Describe data augmentation techniques

Compare custom CNN vs. transfer-learning baselines

Metrics: accuracy, macro-F1, confusion matrix, # parameters

Discuss results vs. Ramadan et al. [2]

Framework: PyTorch · Dataset: InfraredSolarModules (~20k IR images)

Dataset Overview

Dataset properties

InfraredSolarModules — public IR dataset from Raptor Maps
~20,000 thermal images of PV modules
12 classes: 11 defect types + No-Anomaly
Native resolution $\sim 40 \times 24$, upscaled to 224×224 for training
Split: 80 / 10 / 10 train / val / test (stratified, fixed seed)

Defect classes (examples)

Cell, Cell-Multi, Cracking, Hot-Spot, Hot-Spot-Multi, Shadowing, D

Key Data Challenge: Class Imbalance

At the 12-class level: 50% No-Anomaly vs. 50% defects — but defects are highly uneven.
Some classes (~1,900 samples) vs. rarest (~175 samples): order-of-magnitude gap.
Multi-11 task is hardest: every sample is a defect, yet classes remain imbalanced.

Mitigation 1

IR-aware data augmentation (training only)

Mitigation 2

WeightedRandomSampler (balanced batches)

Mitigation 3

Focal Loss ($\gamma=2.0$) for multi-class tasks

Preprocessing Pipeline

Pipeline: Raw IR images → Unsharp Mask sharpening → Label mapping → Augmentation (train only) → Normalize & train

Unsharp Mask filter — enhances low-contrast thermal boundaries (micro-cracks, hot spots, soiling).

Preprocessing runs offline once — not repeated every epoch.

Flexible PVDataset class supports all three classification tasks.

Augmentation applied only to training data to prevent data leakage.

Data Augmentation (Task 2)

Geometric & environmental

Horizontal & vertical flips (50%)

Random 90° rotations

Affine: $\pm 5\%$ translation & scale

Gaussian noise (10% of images)

Random brightness/contrast ($\pm 20\%$)

Regularization & test time

CoarseDropout — masks 1–4 random regions

Forces models to use global context, not a single hotspot pixel

Test-Time Augmentation (TTA): original + h-flip + v-flip + 180° —

Albumentations framework — IR/UAV-motivated transforms

Proposed Model — CBAM Residual CNN

Custom 4-stage residual CNN with CBAM attention in every block

Trained from scratch — no ImageNet pretraining

~4.12M trainable parameters

Channel attention

Re-weights feature channels — amplifies thermal signatures relevant to each defect type

Spatial attention

Focuses on localized hot regions — critical for hotspot, cracking, cell defects

Design rationale: ImageNet features help on coarse tasks; domain-specific attention helps on fine-grained IR defect typing.

Baseline Models — Transfer Learning

Model	Family	Params	Fine-tuning strategy
ViT-B/16	Transformer	14.58M	Last 2 encoder blocks
VGG16	Deep CNN	4.46M	Last 4 feature layers
ResNet50	Residual CNN	16.02M	Entire layer4 block
MobileNetV3-L	Lightweight CNN	2.41M	Last 3 inverted blocks
ProposedModel	CBAM CNN	4.12M	Full network (from scratch)

All models: 224×224 input, ImageNet normalization, task-specific classification head

Training Strategy

Optimizer & loss

Optimizer: AdamW (wd=1e-4), trainable params only

Loss: BCE (binary) · Focal Loss $\gamma=2.0$ + label smoothing 0.1 (multi-class)

Scheduler: linear warmup → cosine annealing

Checkpoint: best model saved on validation macro-F1

Additional choices

Gradient clipping (max norm 1.0)

WeightedRandomSampler for class balance

ProposedModel: longer training (30–40 epochs, lr=3e-4)

GPU memory cleanup between sequential model runs

Results — Binary Classification

Model	Acc (%)	F1 (%)
ViT-B/16	91.55	91.22
VGG16	87.55	86.96
ResNet50	90.85	90.50
MobileNetV3	90.00	89.62
ProposedModel	92.45	92.23

ProposedModel

92.45%

Best accuracy & F1 — with $\sim 3.5\times$ fewer params than ViT

Clean triage task: all five models within ~ 5 points; our CNN wins on effici

Results — 11-Class (Defects Only)

Hardest task: visually similar IR patterns, severe class imbalance, no easy No-Anomaly class

Model	Acc (%)	Macro-F1 (%)	Gap to ProposedModel
ViT-B/16	65.40	61.29	−9.37 F1
ResNet50	64.50	60.30	−10.36 F1
MobileNetV3	62.90	58.92	−11.74 F1
VGG16	55.80	49.71	−20.95 F1
ProposedModel	73.10	70.66	—

Rare classes (Diode-Multi, Soiling) reach $F1 \geq 0.94$ thanks to Focal Loss + weighted sampling

Results — 12-Class (Full Taxonomy)

Model	Acc (%)	Macro-F1 (%)
ViT-B/16	77.45	63.51
ResNet50	72.20	60.08
MobileNetV3	65.80	55.60
VGG16	57.10	45.53
ProposedModel	78.40	68.28

ProposedModel: best accuracy and best macro-F1

ViT leads on accuracy alone by being conservative on No-Anomaly — not better at defect typing

Strong per-class F1 on Diode (0.95), Diode-Multi (1.00), No-Anomaly (0.88), Cracking (0.80)

Overall Comparison

Task	Metric	ViT	VGG	ResNet	MobileNet	Proposed
Binary	Acc / F1	91.6 / 91.2	87.6 / 87.0	90.9 / 90.5	90.0 / 89.6	92.5 / 92.2
Multi-11	Acc / F1	65.4 / 61.3	55.8 / 49.7	64.5 / 60.3	62.9 / 58.9	73.1 / 70.7
Multi-12	Acc / F1	77.5 / 63.5	57.1 / 45.5	72.2 / 60.1	65.8 / 55.6	78.4 / 68.3
# Parameters	—	14.58M	4.46M	16.02M	2.41M	4.12M

ProposedModel wins 5 of 6 metrics. The performance gap widens as task difficulty increases — domain-specific design beats raw ImageNet capacity on fine-grained IR classification.

Comparison with Literature (Task 4)

Source	Params	Binary Acc	Multi-12 Acc	Multi-12 F1
Ramadan et al. [2] — ViT	~86M	98.2%	95.6%	—
Le et al. [8] — ResNet ensemble	~1.5M	94.4%	85.9%	—
Boutana et al. [9] — ViT	~86M	88.0%	~81.0%	—
Our ViT-B/16 (head-tuned)	14.58M	91.6%	77.5%	63.5%
Our ProposedModel	4.12M	92.5%	78.4%	68.3%

Ramadan [2] is SOTA but needs $21\times$ more parameters + aggressive oversampling

Our strict leak-free 80/10/10 split gives conservative but honest estimates

Right training recipe can substitute for raw model capacity on small, biased datasets

Limitations & Future Work

Remaining failure modes

Hot-Spot vs. Hot-Spot-Multi — differ only in # of affected cells (MixUp 0.55)

Cell vs. Cell-Multi — same ambiguity from a single IR frame

Offline-Module over-prediction from aggressive 1/freq sampling

Next steps

MixUp on minority classes

Knowledge distillation from large ViT → ProposedModel

Self-supervised IR pretraining (SimCLR / MAE)

Multi-frame temporal input · INT8 quantization for UAV deployment

Conclusion

Built three PV defect classifiers on thermal IR data using a rigorous, leak-free evaluation protocol.

Our 4.12M-parameter CBAM-CNN is the best macro-F1 model on both multi-class tasks and best overall on binary.

Recipe: CBAM attention + Focal Loss + WeightedRandomSampler + IR-aware augmentation.

Demonstrates that domain-specific design beats generic ImageNet backbones for fine-grained thermal defect typing — while staying deployable on edge/UAV hardware [1].

Thank you — Questions?